



The Age of Zero

Low-Sugar & Low-Calorie Food Industry

Team D

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1. Low-Sugar & Low-Calorie Industry Strategic Recommendations

I. Reason for Industry Selection & Industry Overview

1. Diet Industry — Shift in Perception & Definition and Scale of the Low-Sugar/Low-Calorie Industry

Diet – Low-Sugar & Low-Calorie Industry Overview

The emergence and growth of Low-Sugar & Low-Calorie foods following the revival of the health-pleasure trend.

Diet Industry Definitions and Types

Diet Industry Definition

- All products and services intended for weight loss, maintenance and component control (such as increased muscle mass).

Types of Diet Industry

- **“Low-Sugar & Low-Calorie foods”** with reduced sugars and fats
- A “weight control preparation food” processed with nutrients to replace meals
- “Health functional foods” recognized for their functionality in reducing body fat

Low-Sugar & Low-Calorie Food Definition

“What are low-sugar & low-calorie foods?”

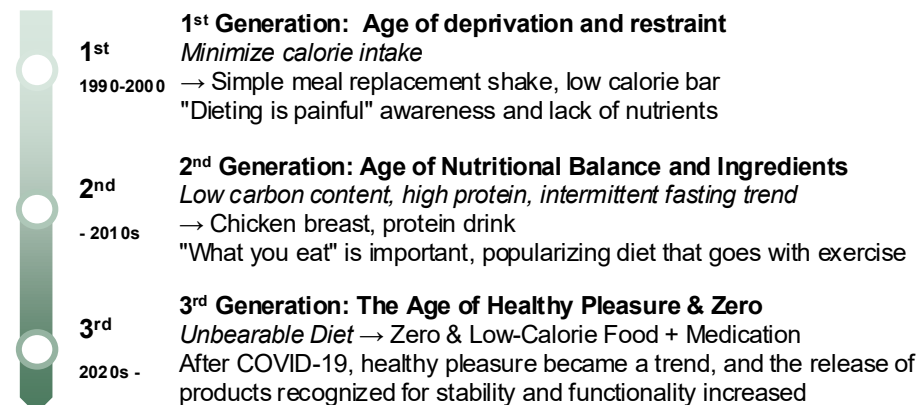
• Zero food

: A product range that meets certain criteria (such as less than 4 kcal per 100 ml) under the Food Sanitation Act.

• Alternative sugar food

: Products with sweetened food additives(replaceable sugars) instead of conventional sugars that dramatically reduce sugar content.

Change in industry perception



Low-Sugar & Low-Calorie Food Market

Domestic Market: Explosive Growth

<Zero-Drink Market>

2018: 1.63 billion KRW

→ 2023: 1,278 trillion KRW(7.8x growth)

- Zero product share in the overall carbonated beverage market by 2025 is approximately 30%

<Zero-Food Appear>

- Carbonated beverage → Extend to all areas of food including bakery, snack.

Global Market: Stable Expansion

<Low-Sugar & Sugar-Free Food Market Size>

2024: 29.02 trillion KRW

→ 2029: 35.28 trillion KRW
(CAGR 3.98%)

<Low-Sugar & Sugar-Free Beverage Market Size>

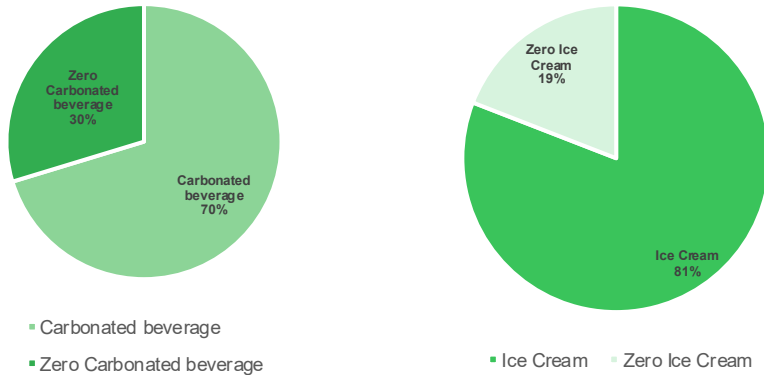
2024년: 5.46 trillion KRW

→ 2033: 17.89 trillion KRW
(CAGR 14.10%)

Low-Sugar & Low-Calorie Industry Growth Profitability Analysis

Low-Sugar & Low-Calorie foods that are rapidly growing in size within the food industry and increasing profitability.

Low-Sugar & Low-Calorie Food Market Size



Increase in Sugar Substitute Imports

Erythritol imports increased by 20.8%
(4,397t in 2024 → 5,291t in 2025)

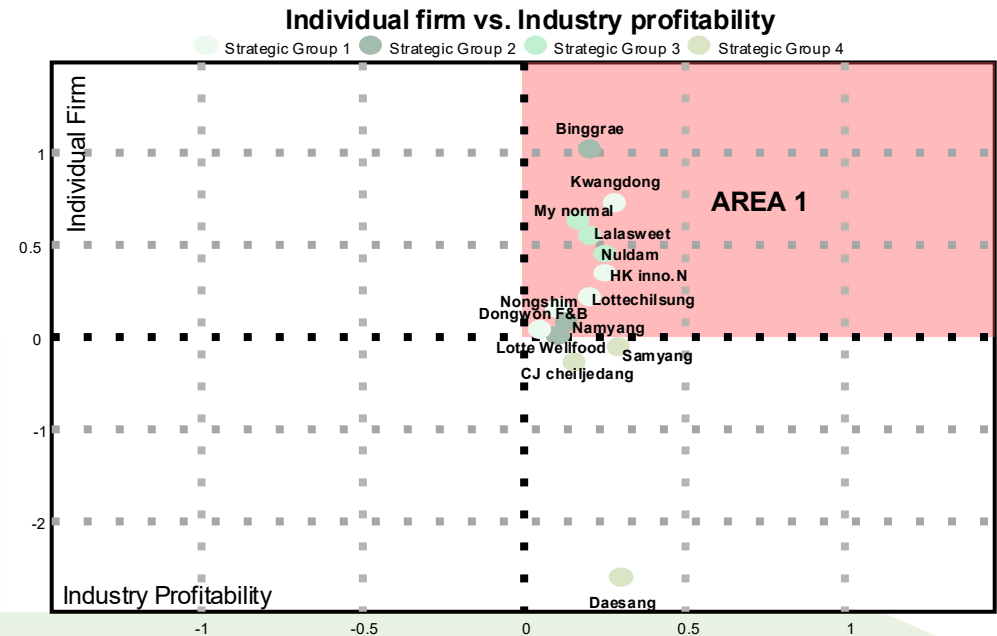
Sucralose imports increased by 27.8%
(241t in 2024 → 308t in 2025)

Individual firm vs. Industry profitability scatter plot

* ROA Calculation Method: Total ROA × Proportion of Zero-Sugar Food

* Mynormal, Nuldam (Unlisted Companies): Utilized proxy indicators (Compared revenue with Lalasweet to estimate ROA)

* Industry Average ROA: Total ROA / Number of Companies



Key Interpretation

The low-sugar & low-calorie industry is experiencing massive growth and high profitability.

II. Industry Attractiveness Analysis by Timeline

1. Past
2. Present
3. Future

Low-Sugar & Low-Calorie Industry - Past Overview

Late 2010s – 2020s: Health demand and policy groundwork emerged, but limited technology, awareness, and regulatory impact held back mainstream adoption.

Executive Summary

- **Pre-2019** — Health and policy interest existed, but weak taste/quality and low awareness held back mainstream adoption
- **~2019** — Next-gen sweeteners (allulose), sugar-reduction policies, and the “healthy pleasure” trend lifted both product competitiveness and demand
- **2021+** — Zero-sugar beverages drove market takeoff; foundation laid for expansion into alcohol, desserts, sauces, and ready meals:



Key Milestones : Past

- 2006-11**
 - Coca-Cola Zero entered the Korean market (2006)
 - Lotte Chilsung launched and discontinued “Chilsung Cider Zero” (2011)
- 2013-19**
 - Obesity tax debate emerged, but legislation failed (2013)
 - 1st Sugar Reduction Plan (2016); Low-Salt & Low-Sugar Initiative (2019)
- 2021**
 - Post-COVID health interest surged
 - “Chilsung Cider Zero” relaunched — breakout hit (2021)
 - Pepsi Zero Sugar (Lime) overtook the market (H2 2021)
- 2022-3**
 - Category expansion: “Saero”, Lotte Wellfood “ZERO” line (2022)
 - Allulose and other next-gen sweeteners gained traction
- 2024-**
 - Expanded CVS shelf space for low-sugar & low-calorie products
 - Zero-sugar RTD teas and functional zero beverages went mainstream
 - Competition intensified; category fragmented

Low-Sugar & Low-Calorie Industry Market Landscape

Domestic Market: Explosive Growth

Domestic Low-Sugar Food Market

- 2021: KRW 210B → 2024: KRW 570B

Domestic Zero-Sugar Beverage Market

- 2016: KRW 90.3B → 2021: KRW 218.9B

Global Market: Steady Expansion

Global Low-Sugar & Sugar-Free Food Market

- 2024: USD 19.35B → 2029: USD 23.52B (CAGR 3.98%)

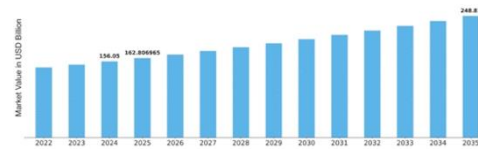
Global Low-Sugar / Sugar-Free Beverage Market

- 2024: USD 3.64B → 2033: USD 11.93B (CAGR 14.10%)

South Korea reduced sugar food & beverages market, 2018-2030 (US\$M)



Low-Calorie Food Market



5 Forces : Past

	Force	Level	Key Drivers
1	Bargaining power of suppliers	Moderate	Sweetener supply existed, but the difference in taste and quality reduces demand
2	Bargaining power of buyers	Low	Limited choices and information asymmetry
3	Threat of new entrants	Low	Low ntry incentives with technology limitations and low consumer awareness
4	Threat of substitute products or services	High	Inferior to conventional sugar in taste and quality
5	Intra-industry Rivalry	Low	Limited competition with a few major firms and specialty players

Past Detailed Analysis (1) - PESTEL & Growth Constraints

Late 2010s – 2020s: Health demand and policy groundwork emerged, but limited technology, awareness, and regulatory impact held back mainstream adoption.

PESTEL Analysis: Past

Political / Policy | Foundations of Sugar-Reduction Policy

- Global sugar taxes & domestic obesity tax debate fueled reduction agenda
- 1st *Sugar Reduction Plan* (2016); *Low-Salt & Low-Sugar Initiative* (2019)
- Focusing on public health, not on industry

Economic | Health Spending Growth & Emerging Niche

- Health functional food market lifted health spending (2015: KRW 1.8T → 2019: KRW 2.95T, CAGR 12.8%)
- Low-sugar products stayed in early-stage, niche territory
- CVS cold-chain and online / D2C channels set the stage for future entry

Social | Rising Health Interest, Limited Awareness

- Interest in health care and weight management increased after COVID-19
- Early signs of healthy pleasure and blood-sugar management trends
- “Zero/low-sugar = bad taste” perception

Technological | Tech Base Set, Quality Lagging

- MFDS approved allulose as a food ingredient (2016)
- Existing sweeteners have unfamiliar aftertaste and low taste similarity
- Low-sugar applications remain limited in desserts, sauces, and ready meals

Environmental | Health Over Sustainability

- Purchase decisions driven by health-weight management, not sustainability
- Sustainability, packaging, carbon — not yet competitive criteria
- Taste, price, calories matter more than ESG

Legal / Regulation | Labeling Reform & Global Tailwinds

- Early stage of domestic sugar labeling and disclosure rules
- U.S. FDA excluded allulose from added-sugar labeling (2019)
- The criteria for zero sugar in Korea are in the pre-elaboration stage

Core Question & Answer

“Despite existing health demand and policy foundations, why did mainstream adoption of the low-sugar & low-calorie industry remain limited in the Past period?”



- Demand & policy existed, but mainstream consumption did not follow
- Weak product competitiveness — driven by taste, price, and quality gaps — kept low-sugar items behind conventional sugar
- Market coverage: niche, health-conscious only

→ **The bottleneck lay in product competitiveness and consumer acceptance, not in demand.**

Growth Constraint Framework



Past Detailed Analysis (2) - 5 Forces Interpretation

Late 2010s – 2020s: Low competition signaled market immaturity and a narrow product portfolio, not high industry attractiveness.

Industry Structure Analysis : Past

Force	Past	Interpretation
Bargaining power of suppliers	Moderate	- Sweetener suppliers existed, but those offering differentiation in taste and quality were limited - Next-generation sweeteners such as allulose were in the early stages of commercialization - Cost and quality concerns kept supplier power moderate
Bargaining power of buyers	Low	- Limited low-sugar product offerings restricted consumer choice - Consumers had little choice because low-sugar options were limited - Narrow consumer base and immature market left buyers with little leverage.
Threat of new entrants	Low	- Allulose and other next-generation sweeteners were still early in commercialization - Low consumer awareness and a limited market size offered weak incentives for new brand entry - Convenience store cold-chain infrastructure and D2C channels were taking shape but had yet to reach full-scale expansion
Threat of substitute products or services	High	- Conventional sugar-based products held the advantage in taste, price, and accessibility. - Aftertaste and unfamiliar mouthfeel of alternative sweeteners limited repurchase incentives for low-sugar products - Low-sugar products functioned more as diet and health-management items than as everyday substitutes
Intra-industry Rivalry	Low	- A limited competitive landscape centered on a few major firms and specialized players. - Insufficient product range, consumer base, and technological maturity placed the industry in a pre-competitive stage - Low competitive intensity resulted not from high profitability, but from low market maturity

Key Interpretation

- Key drag on attractiveness:** substitute threat—inferior to conventional sugar in taste, price, and accessibility
- Low rivalry ≠ profitability:** a signal of market immaturity and narrow product range
- Three unlocks for mainstream growth:** taste similarity, price accessibility, category expansion

Industry Attractiveness : Low ~ Moderate

Technological limitations narrowed the product range and slowed market maturation, leaving attractiveness only modest.

Low-Sugar & Low-Calorie Industry— Present Overview

2020s – Present — Explosive growth in low-sugar & low-calorie foods, with category expansion from beverages into diverse product lines.

Executive Summary

- **2020s+ — Domestic low-sugar & low-calorie market enters full-scale high growth:** 20.1% growth (2022–2023); sales up 2.7× in 3 years
- **Expansion from beverages into non-beverage categories**
— bakery, sauces, desserts, HMR accelerating since 2024
- **Major firms reshape the market:** Large brands expand share and intensify competition across the value chain, from ingredients to finished products



Production Trends of Zero-Sugar Products

- According to MFDS, in 2024, zero-sugar product SKUs more than doubled YoY, with production value rising 20.1%
- The standout shift: growth in non-beverage categories
- Beverages grew 14.4%, while non-beverages surged 109.7% — a far steeper expansion



[MZ세대 추구미 '헬시 플레저'...제과-제빵업계, 저당 트렌드 바람-newsis](#)
[식품안전정책과 보도자료_2024년 식품산업 생산실적 114조 원, 전년 대비 5.8% 증가](#)

Shifting Food Distribution Channels

Core Channel

- **Convenience Stores(CVS)**
: Growth driver for emerging low-sugar / low-calorie categories
- **Online Grocery Platforms**
: Curation exposure supports small-brand visibility

Growth Channel

- **D2C / brand-owned stores**
: Major & niche brands expanding; no commission, full brand control
- **Unmanned “zero-only” stores**
: New retail model targeting health-conscious consumers

Secondary Channel

- **Hypemarkets & chain supermarkets**
: Broad decline due to online shift

5 Forces : Present

	Force	Level	Key Drivers
1	Bargaining power of suppliers	Very High	Oligopolistic control of allulose by mega firms
2	Bargaining power of buyers	Moderate	Online-driven SKU surge and category expansion broadening consumer choice
3	Threat of new entrants	Moderate	Entry barriers raised by major firms' “zero-fication” of mega-brands
4	Threat of substitute products or services	Low	Sugar tax debate weakening price competitiveness of conventional sugar products(= reduced substitute threat)
5	Intra-industry Rivalry	High	Heightened competitive intensity from major firms' full-scale entry

Present Detailed Analysis (1) - PESTEL & Distribution Shift

2020s – Present — Explosive growth in low-sugar & low-calorie foods, with category expansion from beverages into diverse product lines.

PESTEL Analysis: Present

Political / Policy | Government Action to Curb Obesity

- Sugar levy debate gains momentum (2026)
- 6th National Health Promotion Plan (2026–2030) — focus on childhood obesity prevention and management

Economic | Sales Momentum Despite High Inflation

- Inflation + weak consumption + dining-out cost burden → low-sugar sauces rise
- GLP-1 boom → rising demand for complementary low-sugar foods

Social | The Rise of Healthy Pleasure and Low-Sugar Lifestyles

- Continued rise in obesity rates fuels public criticism
- “Healthy pleasure”, blood-sugar management, slow aging, and zero-sugar trends move into the mainstream

Technological | B2B Ingredient Innovation

- Samyang's AI-driven customization service “3S (Smart, Simple, Successful)” : tailored sugar-reduction rates and product specifications to client needs

Environmental | Health Over Sustainability (Continued)

- Taste, price, and calorie competition continue to outweigh ESG differentiation

Legal / Regulation | Ingredient Trust-Building Regulation

- Mandatory alternative-sweetener labeling — effective 2026.1.1. → Boost consumer trust in low-sugar & low-calorie ingredients
- MFDS revises allulose sourcing standards, effectively blocking overseas imports (startup niche brand → cost increase)

Distribution Channel Shifts in the Industry

	Industry-Wide Shifts	Low-Sugar & Low-Calorie Specific
Convenience Stores(CVS)	• Core distribution channel for beverages • investment in fresh-food infrastructure expands fresh-food share	• Core channel for the category • GS25, CU, 7-Eleven all posted double-digit growth in low-sugar SKUs
Hyper-markets	• Broad decline because of the growth of <i>dawn-delivery services</i>	• Expanding shelf space for low-sugar & low-calorie lines
D2C (brand-owned stores)	• Drivers: External-platform commission burden + price-reduction pressure	• Channel dependence varies — healthy-premium lines rely heavily on D2C; snacks & ice cream still depend on CVS (Lotte Wellfood, Binggrae)
Online Grocery Platforms	• Coupang: Rapid delivery, PB expansion • Naver Smart Store: Market-validation stage for startups and small brands • Kurly: Premium / fresh / healthy category hub	• Key online channels and entry channels for niche brands (e.g., Lalasweet) • Small brands gain visibility through curation
Unmanned “zero-only” stores	• (New Format)	• Emerging as a new retail model • Differentiates by narrowly targeting health-conscious consumers

Present Detailed Analysis (2) – Internal Analysis 2026

2020s – Present — Explosive growth in low-sugar & low-calorie foods, with category expansion from beverages into diverse product lines.

Four Key Drivers of Current Growth

Zero Carbonated Saturates

- Zero-sugar carbonated drinks are now an everyday product : consumed by over 8 in 10 consumers (82.1%)
- Growth driven by existing carbonated-drink consumers switching to zero-sugar, rather than new consumer inflow

Samyang's Dominance

- Samyang: #1 in the allulose market ~20% annual growth since beginning allulose mass production (2020)
- Sales surged 30% in 2023
- Pursuing continued growth through overseas expansion and R&D investment

Non-Carbonated Expansion

- **RTD tea market:**
: “Boseong Black Tea Iced Tea Zero” became the market leader in 2024
- **Desserts:**
Zero ice cream posting double-digit growth from 2023 to 2025
- **Ready meals & HMR:**
Dongwon F&B's “Low-Sugar Grain Bowl” sales grew ~83% YoY (2025)
- **Home cooking** goes zero-sugar: natural sweetener purchases +62.0% vs. sugar -24.4%

Major Firms' Line Extensions

- Eases the biggest psychological barrier of low-sugar foods — taste uncertainty
- However, a one-size-fits-all “zero-fication” strategy risks rapid market saturation
- Lack of differentiation among major firms limits sustained customer acquisition

Core Question & Answer

“Amid the all-around growth of low-sugar & low-calorie foods, who actually captures the value?”



Value is concentrated **at both ends** — **upstream** raw-material suppliers (Samyang) and **downstream** major firms with established mega-brands.

• Upstream

- : Quick preempt of allulose raw materials + aggressive R&D and global expansion
- : High growth rate (approximately 20% per year) + Customer (B2B) customized technology development

• Downstream

- : *Binggrae* and *Lotte Wellfood* (ice cream), *Ottogi* (ketchup), *Chung Jung One* (gochujang, red vinegar) are rapidly capturing categories by extending mega-brands into low-sugar lines.
- : New product speed and brand trust work at the same time

Outcome

- : The market is growing, but space for new and small-to-mid brands to enter remains narrow.
- Still, it is premature to declare mega-brands the winners — for new and smaller brands, ideas and preoccupying individual categories are key

Present Detailed Analysis (3) - 5 Forces Interpretation

2020s – Present — Explosive growth in low-sugar & low-calorie foods, with category expansion from beverages into diverse product lines.

Industry Structure Analysis : Present

Force	Present	Interpretation
Bargaining power of suppliers	Very High	Samyang–Daesang duopoly entrenched, with Samyang remaining the #1 allulose market share
Bargaining power of buyers	Moderate	- Low price sensitivity in the market → consumers willing to pay for better ingredients and taste on par with conventional products, rather than the lowest price - B2C) Online distribution drives SKU growth and category expansion → increased consumer choice - B2B) Partial bargaining power (shelf-placement authority, commission and price-cut pressure)
Threat of new entrants	Moderate	- Major firms' line-extension strategy makes entry barriers — difficult to match the same speed and quality - However, new entrants with differentiated positioning can still capture competitive advantage
Threat of substitute products or services	Low	- Sugar levy debate → weakens price competitiveness of conventional sugar products - Mandatory alternative-sweetener labeling: eases the trust issues around low-sugar & low-calorie products
Intra-industry Rivalry	High	- Zero-sugar carbonated drinks are effectively saturated → market-share battle - Major firms (mega-brand low-sugar line extensions) vs. niche brands (ideas & digital marketing) → high rivalry overall, but dispersed across diversifying categories

Key Interpretation

- Mega firms** hold a **structural advantage**, but **niche brands' creative ideas** serve as a **powerful differentiation lever**
- All-around expansion**: the competitive arena has shifted to “early occupation of categories”

Industry Attractiveness : High

Demand-side forces (preferences, regulation, GLP-1) are aligning to boost industry attractiveness

Low-Sugar & Low-Calorie Industry – Future Overview

The Next Decade: Low-Sugar & Low-Calorie Moves Beyond a Trend to Become the F&B Industry Standard.

Executive Summary

- **Ongoing growth in both domestic and global markets**
: An attractive market sustaining 7%+ CAGR through the 2030s
- **Low-sugar & low-calorie as a mainstream culture**
: “Healthy pleasure” trend remains still throughout the 2020s
- **Industry restructured through technological and regulatory progress**
: A structure favoring firms with proprietary technology; IT integration becomes inevitable



Key Milestones & Future Value



PESTEL Analysis: Future

Political / Policy | Full-Scale Introduction of Sugar Tax

- Rising demand for alternative sweeteners following the introduction of a sugar levy
- Alternative-sweetener ingredient producers benefitting from sugar tax

Economic | Rising Profits from Market Growth

- Explosive global market growth secures a broad consumer base
- Industry development lowers production costs, reducing expenses
- Industry attractiveness rises over the mid-to-long term

Social | Long-Term Mainstreaming of the Industry

- Health-consumption trends entrench; low-sugar & low-calorie consumption becomes a culture
- Rising chronic-disease risk further drives health-conscious consumption

Technological | AI & Platform Convergence

- Advancement of AI-based alternative-sweetener formulation technology
- Close integration with data-driven, hyper-personalized diet platforms
- Tech progress shifts the industry paradigm mid-to-long term

Environmental | Sustainable Production Solutions

- Advances in alternative sweetener protein production technology
- Cuts land & water use by 60–90% compared to the conventional sugar production

Legal / Regulation | Only R&C-Strong Firms Survive

- Mandatory sweetener labeling + personal health data laws → only firms with superiority in alternative-sweetener formulation, ingredient safety, and data security survive — stabilizing the industry as a whole

Future Expectation – Short Term

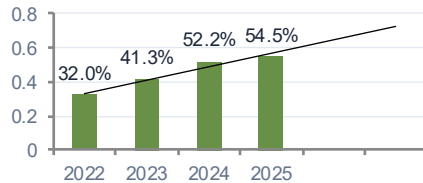
2026–2028: The Battle to Claim “Zero” Territory — A Crowded Market of Firms and Products.

Intensifying Competition in the Domestic Market

Health trends sustained

- “Healthy pleasure” and slow-aging trends continue through the late 2020s
- Zero begins to go mainstream

Zero Share of Carbonated Sales (%)



Within-market expansion

- Zero-category expansion brings more firms into the market
- Within-brand portfolio diversification — e.g., beverages → snacks → sauces and beyond
- As of 2023, beverages generated the largest profits in the domestic low-sugar F&B market, but the snack segment is also expected to surge

Continued major-firm entry

Shrinking entry intervals among major firms

- **2020** Dongwon Home Food — launched “Vivid Kitchen”, a meal-management brand
- **2022** Lotte Wellfood — launched “ZERO”, a sugar-free brand
- **2025** Binggrae — launched “Deep & Low”, a low-sugar ice cream line
- **2025** Ottogi — launched “Light & Joy”, an integrated low-sugar-calorie brand
- **2025** Daesang — introduced “LOWTAG”, a low-sugar & low-calorie emblem

D2C SMEs disadvantaged

- Unlike SMEs taking **months** per product, major firms leverage existing infrastructure to push multiple SKUs into distribution **at once**
- Threats from major firms **undercutting** SMEs with same-factory, same-recipe products
- Mass emergence of “**copycat**” firms imitating D2C marketing strategies wholesale
- Small D2C firms focused solely on low-sugar & low-calorie are disadvantaged
- Strategy must shift from speed to **quality and R&D**

Short-Term Impact of Macro Variables

- **Ongoing controversy over the efficacy of alternative sweeteners**
 - : Past controversies over the carcinogenic potential of certain synthetic sweeteners
 - : Sugar alcohols disrupting gut microbiome balance
 - : Negative effects on taste perception of sweetness
- **WHO’s statement**

“High-intensity sweeteners such as *aspartame* are not effective in reducing body fat over the long term and may increase the risk of type 2 diabetes and cardiovascular disease.”

5 Forces Analysis : Short-term Future

- Excluding beverages, moderate competition remains — industry attractiveness still high

	Force	Level	Key Drivers
1	Bargaining power of suppliers	Very High	Concentration toward a few suppliers holding tech and regulatory approval
2	Bargaining power of buyers	Moderate	Smart consumers buying based on nutritional content
3	Threat of new entrants	Moderate	Major firms with capital and distribution entering the market
4	Threat of substitute products or services	Moderate	Low substitute threat as GLP-1 plays a complementary role
5	Intra-industry Rivalry	High	Intensifying share competition led by major firms amid low-sugar trend expansion

Future Expectation – Short Term

2026–2028: The Battle to Claim “Zero” Territory — A Crowded Market of Firms and Products.

Core Question & Answer (1)

“Zero’ consumption taking root as a culture vs. ongoing controversy over alternative sweeteners — which force will drive the whole industry?”



New information on alternative sweetener **efficacy/safety/side effects** triggers temporary consumption pullback, but **long-term impact is minimal**.

- 1) Low market share of *aspartame*, *saccharin*, etc. makes the category relatively unaffected by synthetic-sweetener controversies

Domestic Sales Share of Food Additives

Sweeteners	Share
Aspartame	0.05%
Sodium Saccharin	0.17%
Enzymatically Modified Stevia	1.16%
D-Sorbitol Solution	1.05%

- 2) Even nutrition-conscious consumers tend to choose alternative sweeteners over sugar (sugar-avoidance > side-effect concerns)
⇒ Firms should avoid overreacting to short-term volatility and proactively signal that their products are clear of controversy

Core Question & Answer (2)

“Will GLP-1 anti-obesity drugs continue to function as complements to diet foods?”



The complementary relationship will persist.

- GLP-1 drugs only support temporary weight loss; permanent effects do not come from a few uses
- Future demand for low-sugar & low-calorie foods will be driven by weight maintenance after weight loss, not weight loss itself
- As oral GLP-1 drugs commercialize alongside the existing injectable form, the GLP-1 user base will expand
⇒ This drives growth in the population habitually consuming low-sugar & low-calorie foods

Future Expectation – Middle Term

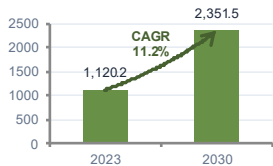
2028–2031: Unpredictable Market Restructuring
— **Sugar Tax Debate and Tech/Infrastructure-Driven Global Expansion.**

Sustained Market Expansion

Positive outlook

Total revenue of the domestic low-sugar F&B market is projected to roughly **double** over seven years (2023–2030)

Domestic Low-Sugar F&B Market Revenue (KRW billions)



Accelerating Global Entry

Dongwon Home Food

- promoted “Vivid Kitchen” products at Expo West 2025 (Anaheim, California)

Lotte Wellfood

- “ZERO” exported to 13 countries (+a)
- ; Q1 2025 export sales of the ZERO brand grew 4× YoY

Samyang

- plans to double the overseas share of its specialty business, including allulose, by 2030

Restructuring of the Low-Sugar Market

• Rising demand for natural alternative sweeteners

: Precision fermentation drives innovation in natural high-intensity sweeteners
+ Distrust of synthetic sweeteners ⇒ Sustainable, natural solutions

• Intensifying competition among Niche brands

: Market penetration by domestic and global majors + Regulation of misleading advertising + Diversifying diet trends ⇒ New brand strategies become essential

• Shift toward tech-driven business structures

: Securing technology becomes critical — Samyang’s “3S Solution”, CJ’s “Sugar Reduction Modulation” technology, AI for alt-sweetener formulation, ingredient patents ⇒ Investment in R&D and tech integration

Scenarios Depending on the Domestic Sugar Tax

Case1. No Sugar Tax

- Market continues to grow — driven by category & global expansion
- Industry still favors alt-sweetener producers and low-sugar food firms, even without a sugar tax
- However, sugar substitution remains non-mandatory

Case 2. Sugar Tax Implemented

- Two competing views on the expected impact:
 - 1) Deliver major gains to allulose and other alternative-sweetener producers
 - 2) The “zero” trend has gone mainstream + alternative-sweetener adoption already substantial
→ limited impact from a sugar tax

5 Forces Analysis: Middle-term future

- Intensified Threats and Competition, with Gradually Declining Attractiveness

	Force	Level	Key Drivers
1	Bargaining power of suppliers	High	Continued market dominance by ingredient suppliers
2	Bargaining power of buyers	High	Rising consumer health standards and pressure for higher product quality
3	Threat of new entrants	High	Greater transparency in technological information
4	Threat of substitute products or services	Moderate	Only internal substitution exists
5	Intra-industry Rivalry	Very High	Sustained B2C competition + intensifying B2B leadership battle among ingredient suppliers

Future Expectation – Long Term

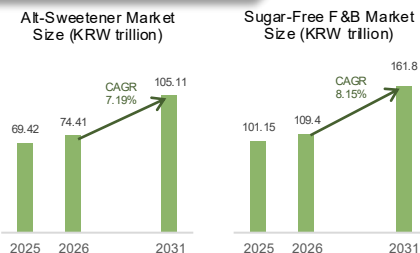
2031–2036: The “Zero” New Normal

— Global Hegemony Competition and Platform-Driven Industry Restructuring.

Building a Sustainable Healthcare Ecosystem

Growth in Global Market Size

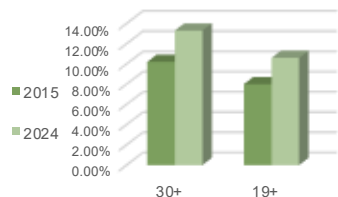
- Global sugar-free F&B, low-calorie F&B, and alternative-sweetener markets all projected to grow ~1.4–1.5× in size
- Firms move beyond domestic markets to compete globally — and grow at a larger scale
- Active, large-scale profit generation across broader global markets



Rising Prevalence of Diabetes and Obesity

- Higher adult-disease risk driven by modern lifestyles and shifting social structures
- Domestic diabetes prevalence (ages 19+): 8.0% → 10.6% (2015–2024)
- Domestic obesity prevalence (2015–2024): +10pp among those in their 20s, +5pp among those in their 30s
- Patient growth concentrated in younger demographics → sustained demand for dietary management → expanding market size

Domestic Diabetes Prevalence



Even with GLP-1 anti-obesity drugs commercialized, the underlying threat of adult diseases persists

Rising disease prevalence signals more than patient growth
 → Unhealthy lifestyles entrenched among younger demographics
 → Even with anti-obesity drugs reducing disease prevalence, ongoing management remains essential
 → Diet foods persist as the simplest, sustainable solution

Low-Sugar & Low-Calorie as Lifestyle

Mainstreaming of Multi-Functional Hybrid Foods

- Low-sugar & low-calorie = the default
- Hybrid foods with added nutrients/functional ingredients drive long-term value
- Shift from low-spec to high-spec foods

Diversifying and Segmenting Target Consumers

- Firms actively pursue niches as the market grows
- Existing brands broaden portfolios into segments like senior low-sugar foods
- New brands emerge with expertise in specific target groups

Hyper-Personalized Services Through Digital Channels

- Growth in AI- and data-driven diet consumption
- Convergence of diet foods with technology (e.g., FoodLens)
- Emergence of platform firms specializing in low-sugar & low-calorie foods
- Diet apps: CAGR 6.4% (2018–2027)

5 Forces Analysis: Long-term future

- Surviving only with Differentiation — Low~Moderate Attractiveness

	Force	Level	Key Drivers
1	Bargaining power of suppliers	Moderate	Decreased scarcity as ingredient production becomes mainstream; platforms gain influence
2	Bargaining power of buyers	Very High	Rising demand for hyper-personalized services
3	Threat of new entrants	High	Alt-sweetener formulations become commoditized; IT and platform firms enter as new players
4	Threat of substitute products or services	High	As products become everyday staples, substitution threat declines
5	Intra-industry Rivalry	Very High	Unlimited competition with multinational F&B firms worldwide

Future Expectation – 5 Forces Flow

2026–2036: Industry Attractiveness Gradually Declines

Despite the Category Becoming Indispensable and Supplier Power Easing — Due to the Industry's Inherent Complexity.

Industry Structure Analysis : Future Expectation

Force	Short Term (2026-28)	Middle Term (2028-31)	Long Term (2031-36)
Bargaining power of suppliers	Very High Hard-to-acquire licenses and technologies owned by only a few → effectively a monopoly/oligopoly	High Continued market dominance through core ingredient technology and infrastructure; supplier bargaining power may rise further if a sugar tax is introduced	Moderate Reduced scarcity in ingredient production as transparency expands; growing influence of data-supplying platforms
Bargaining power of buyers	Moderate Active search for nutritional labeling information; growing range of product choices	High Rising consumer health standards; pressure for higher quality driven by greater demand for natural sweeteners	Very High Customized products/services preferred over mass production; integration with data-driven hyper-personalization in demand
Threat of new entrants	Moderate Entry concentrated among major firms due to capital and distribution barriers; higher survival barriers for small brands	High Entry barriers remain due to advanced R&D infrastructure requirements; however, growing transparency in formulation technology lowers entry hurdles	High IT and platform firms (AI, FoodLens, etc.) enter directly or via “borrow” strategies, threatening traditional business models
Threat of substitute products or services	Moderate Complementary relationship with GLP-1 anti-obesity drugs; reduced substitute threat as the zero-beverage segment matures	Moderate High internal substitution: synthetic-sweetener avoidance + tech advances drive shift to precision-fermented natural high-intensity sweeteners	Low “Zero” becomes the default and irreplaceable; high substitution threat <i>within</i> sub-segments of the industry
Intra-industry Rivalry	High Expansion of the low-sugar & low-calorie trend; intensifying competition with varying intensity across categories	Very High Sustained B2C competition drives diversification of D2C brand strategies; B2B battle for ingredient hegemony	Very High Domestic market broadly matures; competition with global multinationals

Key Interpretation

1. Follow evolving regulations and diet trends from a consumer's perspective
2. Secure core competence through sustained R&D investment and the adoption of emerging technologies.
3. Broaden perspective to encompass global markets and adjacent segments of the diet industry.

Ultimate Implications

1. platform integration
2. target expansion

III. Strategic Group

1. Strategic Group Overview
2. Group 1: Brand Extension — Zero Beverages
3. Group 2: Brand Extension — Low-Sugar Desserts/Snacks
4. Group 3: D2C Niche Pioneers — Low-Sugar Desserts/Snacks
5. Group 4: Ingredient Weaponization — Alternative Sweeteners

Strategic Group Overview

The low-sugar / zero-sugar market is composed of four strategic groups: Brand Extension (Beverages), Brand Extension (Desserts/Snacks), D2C Niche Pioneers, and Ingredient Weaponization.

Four Strategic Groups

1. Brand Extension — Zero Beverages

- Core assets: high brand recognition of established mega-brands and powerful nationwide distribution networks
- Expands zero-product lineups at low risk and high speed within the mature beverage market



2. Brand Extension — Low-Sugar Desserts/Snacks

- Combines existing brand equity with in-house manufacturing infrastructure and R&D capabilities
- Leverages existing mega-brand products to significantly expand the frozen-dessert and dairy markets pioneered by Strategic Group 3 startups



3. D2C Niche Pioneers — Low-Sugar Desserts/Snacks

- Grows on the strength of First Mover positioning and powerful fandom-based marketing
- Defines a new market by reformulating high calories desserts and snacks (“guilty pleasures”) into low-sugar versions



4. Ingredient Weaponization — Alternative Sweeteners

- Holds patented technology for alternative sweeteners and global regulatory-approval barriers as core assets
- Dominates the entire value chain, from ingredient supply (B2B) to finished products (B2C)



Group 1: Brand Extension — Zero Beverages

Strategic Group 1 consists of firms that have entered the market by adding zero-product lines to their existing beverage brands and distribution networks.

Executive Summary

- **Core Resources**
Brand equity + manufacturing infrastructure + R&D capabilities
- **Strategic Characteristics**
Line extension by attaching “zero / sugar-free” tags to existing mega-brands
A mature market nearing saturation, as most major players have already entered
Brand power serves as the core source of competitive advantage
- **Distribution Network**
Leverages existing beverage distribution channels (CVS, hypermarkets, online) without the need for new channel development for zero-product lines
- **Profitability**
Minimal additional capital investment required — only the sweetener is swapped within existing beverage lines and distribution networks
Growth is slowing as the market approaches saturation
Intensifying price competition among brands may put pressure on margins

Companies and Key Products

Lotte Chilsung

Pepsi Zero Sugar,
Chilsung Cider Zero,
Saero (zero-sugar soju)

Dongwon F&B

Boseong Black Tea Iced
Tea Zero,
Coolpis Talk Zero

Nongshim

Welch's Zero

Kwangdong

Vita 500 Zero

HK inno.N

Tealog

VRIO Analysis

	Factor	Result	Evidence
1	V: Value	✓	They put value of zero sugar in the existing brand that consumers love
2	R: Rarity	✗	Any major firm can do the same
3	I: Imitability	✗	Low technological barriers
4	O: Organized	✓	Great quality of existing manufacturing system and distribution networks

5 Forces : Strategic Group 1

- Only overwhelming-scale firms like Lotte Chilsung can survive, as just one of the five forces (supplier power) works in their favor.

	Force	Level	Key Drivers
1	Bargaining power of suppliers	Low	Diverse supplier base of commodity sweeteners Their brand equity and scale has a significant power.
2	Bargaining power of buyers	High	Brand recognition matters, but low switching costs between zero-sugar carbonated drinks allow consumers to easily choose alternatives
3	Threat of new entrants	High	Low technological barriers — adding a zero-product line is easy for any existing beverage brand
4	Threat of substitute products or services	Moderate	Substitutes include naturally sugar-free health drinks (water, sparkling water, tea) and indulgence beverages (coffee, juice)
5	Intra-industry Rivalry	Very High	Extreme competition as existing beverage firms enter with similar strategies

Group 2: Brand Extension — Low-Sugar Desserts/Snacks

Strategic Group 2 consists of firms that have entered the new low-sugar dessert category by mobilizing their manufacturing infrastructure and brand equity.

Executive Summary

- **Core Resources**
Brand equity + manufacturing infrastructure + R&D capabilities
- **Strategic Characteristics**
Same line-extension strategy as Group 1, but applied to a still-growing category
Replicating taste, texture, and shelf-stability in solid foods is more technically demanding than in beverages, but entry remains feasible thanks to existing manufacturing infrastructure and R&D capabilities
A structure where major firms entered after Lala Sweet pioneered the category
- **Distribution Network**
Leverages existing offline distribution channels (CVS, hypermarkets)
- **Profitability**
Low marginal cost — no additional capital investment needed, as existing infrastructure is reused
Lotte Wellfood's "ZERO" brand grew over 120% YoY after launch, generating roughly KRW 40B in sales



Companies and Key Products

Lotte Wellfood
ZERO brand
(cookies, jelly, etc.),
Jaws Bar·Screw
Bar·Watermelon Bar
Okcal

Binggrae
Heat-Buster Zero
Saenggyul
Taenggyul Zero
Yoplait Zero (sugar-free)

Namyang
Bulgari Zero,
French Café
Roastery Zero Sugar

VRIO Analysis

	Factor	Result	Evidence
1	V: Value	✓	Existing brand trust enables consumer adoption of zero-product lines with no switching costs
2	R: Rarity	△	Many large food firms could hold these capabilities, but few also have frozen-dessert and dairy manufacturing capacity
3	I: Imitability	△	Short-term imitation is difficult due to required capex and R&D timelines — but feasible long term
4	O: Organized	✓	Immediately deployable through existing distribution networks and sales organizations

5 Forces : Strategic Group 2

- More attractive than Group 1, but substitute threats and intra-group competition remain high

	Force	Level	Key Drivers
1	Bargaining power of suppliers	Low	Diverse supplier base of commodity sweeteners; major dessert/snack firms exert significant influence through purchasing scale and brand recognition
2	Bargaining power of buyers	Moderate	Slight switching costs from greater product differentiation than beverages Brand lock-in through appeal to fans of established frozen-dessert and confectionery brands
3	Threat of new entrants	Moderate	Moderate technological entry barriers, as replicating taste, texture, and shelf-stability is more demanding than in beverages
4	Threat of substitute products or services	High	Group 3 products serve as direct substitutes
5	Intra-industry Rivalry	High	Lotte Wellfood and Binggrae are locked in a tight competition in the frozen-dessert market

Group 3: D2C Niche Pioneers — Low-Sugar Desserts/Snacks

Strategic Group 3 is a startup-driven type that has claimed the low-sugar dessert category through strong marketing and fandom.

Executive Summary

- **Core Resources**
Category First Mover positioning + fandom
- **Strategic Characteristics**
Concentrated on branding and marketing to claim first-mover position in the “low-sugar guilty pleasures” category, which major firms had overlooked
Category overlaps with Group 2, but the key difference lies in building everything from scratch without inherited assets
- **Distribution Network**
Online & D2C: Uses brand-owned stores as the lead channel, with major e-commerce platforms like Coupang and Kurly as core channels
Offline: Convenience stores serve as the springboard for offline expansion
Global: Expanding overseas, including entry into ~100 retail outlets in Vietnam
Captures channels optimized for the niche market by utilizing data from SNS fandom engagement, rather than relying on the traditional mass-distribution model used by major firms
- **Profitability**
Low fixed costs via outsourced manufacturing — but high variable costs from OEM unit prices and SNS marketing make cost control a structural weakness
Margin defense becomes critical if major firms launch similar products at lower prices
Lala Sweet has sustained 50%+ annual growth without external investment, suggesting a healthy cash-flow structure operated entirely on its own revenue

Companies and Key Products

Lala Sweet

Low-sugar chocolate bars·ice cream·popcorn·macarons

My Normal

Low-sugar snacks, low-sugar sauces



Nuldam

Low-sugar desserts

VRIO Analysis

	Factor	Result	Evidence
1	V: Value	✓	Defined the low-sugar dessert category before major firms entered, capturing consumer awareness first
2	R: Rarity	✓	Lala Sweet currently holds the sole Category First Mover position
3	I: Imitability	△	Products can be replicated by major firms; fandom and brand storytelling cannot
4	O: Organized	✓	Vertical integration (production to distribution) + in-house SNS marketing → rapid new-product responses

5 Forces : Strategic Group 3

- High initial costs suppress supplier/buyer bargaining power and entrant threats — a stable market

	Force	Level	Key Drivers
1	Bargaining power of suppliers	High	Reliance on OEM over in-house factories → strong manufacturer bargaining power, weak cost control
2	Bargaining power of buyers	Low	First-mover category definition + fandom branding → high consumer switching costs
3	Threat of new entrants	High	Disadvantaged in capital and economies of scale as Group 2's majors enter
4	Threat of substitute products or services	Moderate	Group 2's zero-products act as substitutes, but full substitution is difficult due to brand storytelling and fandom
5	Intra-industry Rivalry	Moderate	Intra-group competition still moderate — Lala Sweet dominates while others occupy niches

Group 4: Ingredient Weaponization — Alternative Sweeteners

Strategic Group 4 consists of firms that supply alternative-sweetener ingredients to both businesses and consumers, leveraging patents, process technology, infrastructure, and global regulatory approvals.

Executive Summary

• Core Resources

Proprietary enzyme and process technology + global regulatory approvals + large-scale production infra

• Strategic Characteristics

1. Vertical Integration: Upstream supply to Groups 1–3 + downstream in-house brands selling alt-sweetener products to consumers

2. Integrated Brand Strategy: Strengthens awareness by launching dedicated brands that encompass the alternative-sweetener category (e.g., Daesang's "Sweetber", Samyang's "Q-One TrueSweet")

• Distribution Network

1. B2B: Direct supply of core ingredients to major domestic and global food & beverage firms, including Lotte Chilsung, Dong-A Otsuka, and HiteJinro

2. B2C: Utilizes major firms' established distribution networks — supermarkets, e-commerce platforms (Coupang, Kurly, etc.), and brand-owned online stores

3. Global: Ingredient and finished-product exports to the U.S. (North America), China, Japan, Southeast Asia, and other markets, supported by regulatory approvals

• Profitability

High growth: Sharp revenue growth — allulose sales up ~60% YoY on the back of alt-sweetener market expansion

High entry barriers: Patents, regulatory approvals, and large-scale capex limit new entry, enabling stable margin capture



Companies and Key Products

Samyang

Q-One TrueSweet (B2C), Nexweet (B2B global)
→ Strongest B2B expertise

Daesang

Chung Jung One Allulose
B2B + B2C in parallel

CJ CheilJedang

BIO-division alt-sweetener ingredients
B2B + B2C in parallel

VRIO Analysis

	Factor	Result	Evidence
1	V: Value	✓	Serves as the supplier base for Groups 1, 2, and 3 — upstream ingredient supply with high supplier bargaining power
2	R: Rarity	✓	Only 2–3 domestic producers; very few globally
3	I: Imitability	Δ	First-mover lock-in on enzyme patents + mass-production know-how + global regulatory approvals
4	O: Organized	✓	Expanding into B2B sales organizations, global subsidiaries, and AI-based solutions

5 Forces : Strategic Group 4

- High initial costs suppress supplier/buyer bargaining power and entrant threats — a stable market

	Force	Level	Key Drivers
1	Bargaining power of suppliers	Low	Low external supplier dependence due to in-house technology and process control
2	Bargaining power of buyers	Low	With only 2 domestic allulose producers, buyer firms are disadvantaged in price negotiations
3	Threat of new entrants	Low	Triple barrier — patents, process know-how, and global regulatory approvals — means new entry requires several years
4	Threat of substitute products or services	Moderate	Multiple alternative sweeteners and, fundamentally, sugar itself serve as strong substitutes
5	Intra-industry Rivalry	Moderate	A small oligopolistic structure, but competition is expanding around B2B client lock-in and B2C marketing/brand recognition

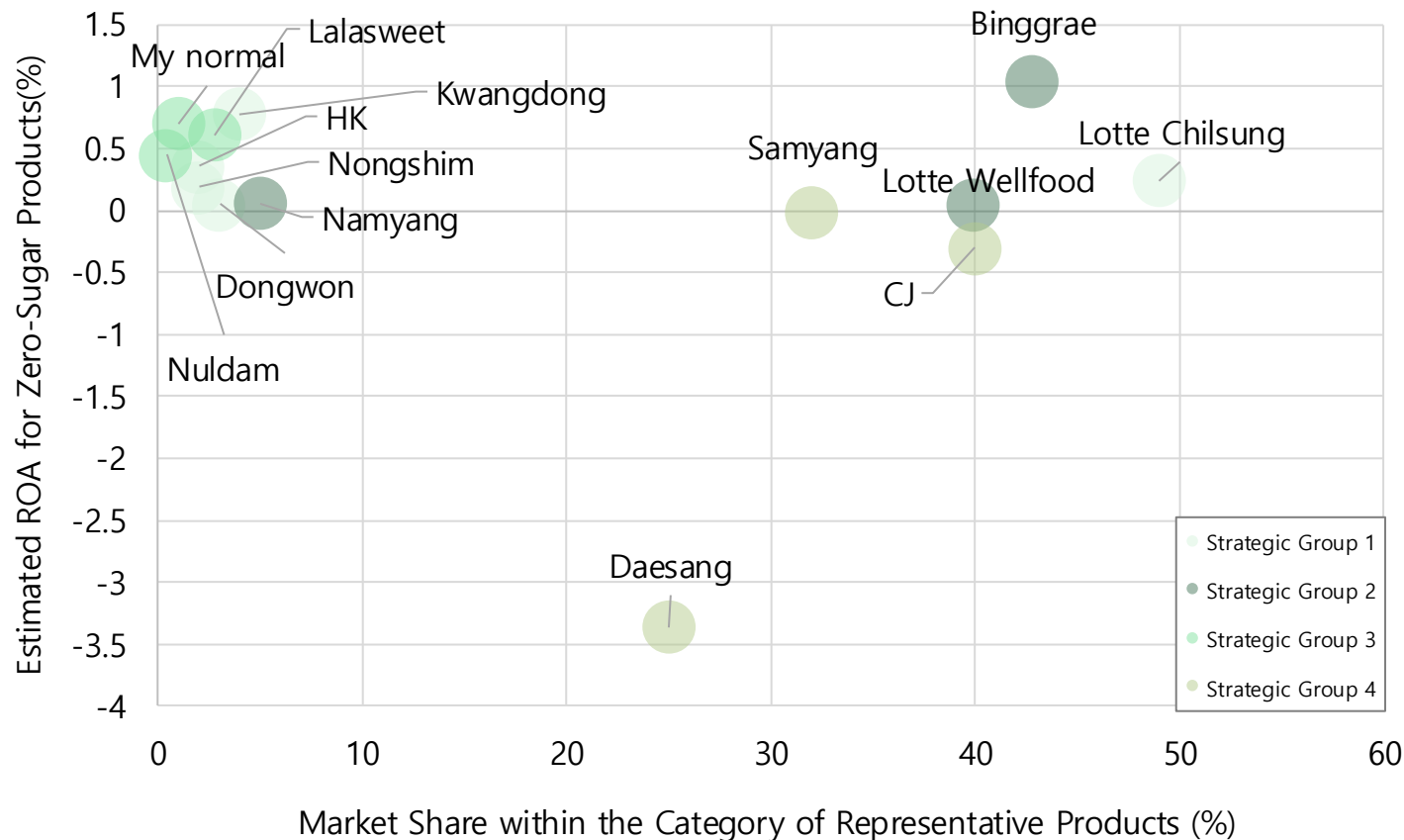
IV. Winner & Winner Factor Analysis

1. Corporate Positioning by Strategic Group – Estimated ROA & Market Share for Zero-Sugar Products
2. By Strategy Group Profit Frontier
3. Winner Company Overview
4. Group 1 Winner: Lotte Chilsung
5. Group 2 Winner: Binggrae
6. Group 3 Winner: Lalasweet
7. Group 4 Winner: Samyang

Corporate Positioning by Strategic Group – Estimated ROA & Market Share for Zero-Sugar Products

Each company is positioned by estimated ROA of zero-sugar products and its market share across the four strategic groups.

Corporate Positioning Map within the Low-Sugar & Low-Calorie Food Industry



Why did we identify winners by strategic group rather than an overall winner?



Strategic groups ①~④ have fundamentally different business models, profit structures, and competitive indicators, making it difficult to directly compare between groups.

* Dongwon, Nongshim, Kwangdong, HK, Namyang, My Normal, and Nuldam are estimated based on sales volume and market structure as public market share data have not been confirmed.

* Estimated ROA for Zero-Sugar Products = Total ROA × Proportion of Zero-Sugar Products

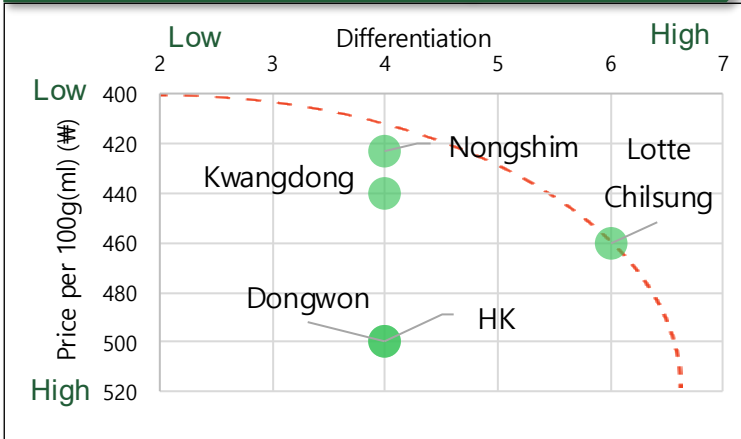
* Group 4: Includes both B2B and B2C revenue

Profit Frontier by Strategic Groups

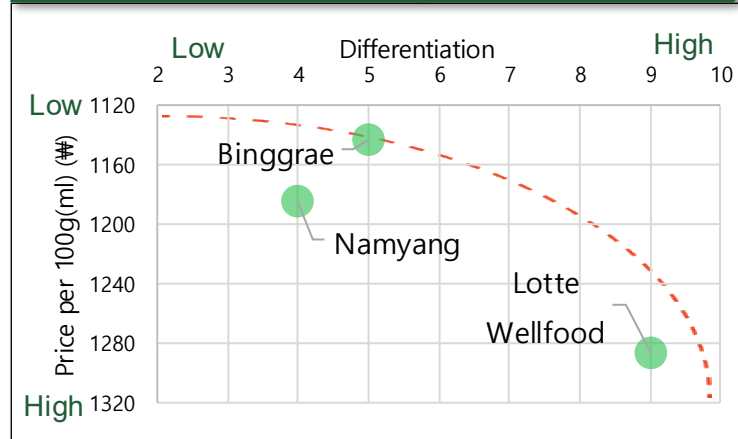
Compare the positions of companies in each strategic group with the Profit Frontier map, centering on the price and differentiation index.

4 Strategic Groups' Profit Frontier

1: Brand Extension — Zero Beverages



2: Brand Extension — Low-Sugar Desserts/Snacks

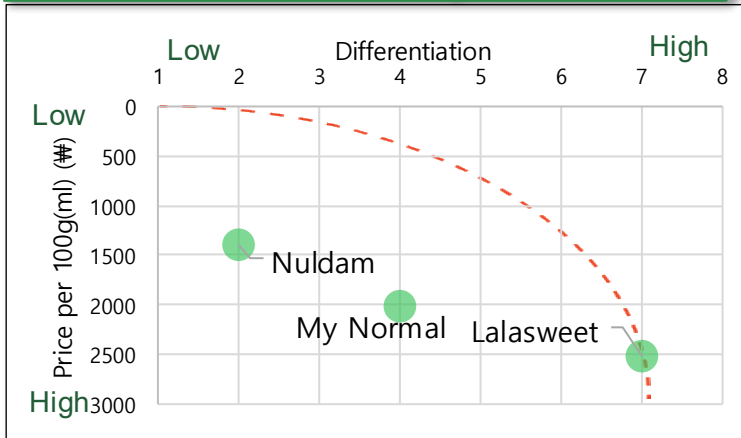


Price Selection Criteria

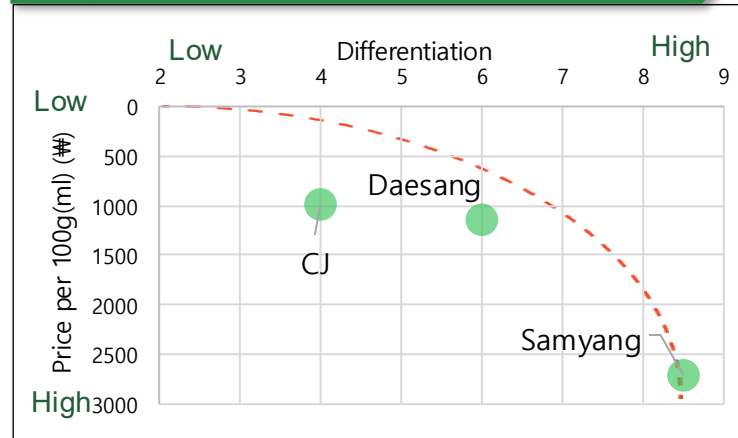
Price per 100g (ml) of representative products similar to other enterprise products in the strategy group

Differentiation Index Calculation Criteria

3: D2C Niche Pioneers — Low-Sugar Desserts/Snacks



4: Ingredient Weaponization — Alternative Sweeteners



① **Product Diversity**
(Number of SKUs of Zero & Low-Sugar Products)
1~3 Types: 1 point / 4~6 Types: 2 points / 7~10 Types: 3 points / 11~15 Types: 4 points / 16 Types: 5 points

② **Technological Capability / Barriers to Entry**
- No factory or technology of its own: 1 point
- Self R&D : 2 points
- Own a factory: 3 points
- Possession of patent or proprietary technology: 4 points
- Patent + Global Licensing + Independent Process: 5 points

Winner Company Overview

Lotte Chilsung, Binggrae, Lalasweet, and Samyang are the winners of each strategic group, and we would like to conduct an analysis of each company.

4 Strategic Group

Group 1 : Lotte Chilsung

- Leveraged the 70+ year brand equity of Chilsung Cider to establish a strong presence in the zero-sugar beverage market
- Dominated the domestic zero-drink market with a 49% market share in 2024
- Increased the share of zero-sugar beverages in total soft drink sales from 12% to 30% since 2021

Group 2 : Binggrae

- Recorded the highest estimated ROA of zero beverages at 1.04%
- After acquisition of Haitai, strengthened its position in the ice cream market, securing 43% of combined market share
- Expanded zero-sugar offerings with established bestsellers
- Operates an extensive distribution networks including local supermarkets, hypermarkets, and convenience stores nationwide

Group 3 : Lalasweet

- Ranked No.4 ice cream manufacturer in Korea, and No.1 in both sales revenue and unit sales within the CU convenient store, outperforming major brands
- Successfully captured the target segment with a clear concept of 'low-sugar', 'sugar-free', and 'low-calorie', with premium pricing over KRW 3,000
- Shifted the overall ice cream market toward zero-sugar trends

Group 4 : Samyang

- Achieved Korea's first mass production of allulose, with No. 1 market share in domestic allulose market
- Completed Korea's largest allulose production facility in Ulsan
- Supplies alternative sugar ingredients to Group 1–3 companies alongside B2C products like 'Q.one'
- Enzyme technology patents, mass-production process know-how, and global regulatory approvals (FDA, GRAS) secure a sustainable competitive advantage

Group 1 Winner: Lotte Chilsung

Lotte Chilsung achieved a 49% market share for zero drinks based on its accumulated mega-brand and overwhelming distribution dominance over decades.

Executive Summary

Item	Details
Zero Product Lineup	Chilsung Cider Zero, Pepsi Zero Sugar (Lime/Mango/Pineapple Apple), Saero (Soju), Milkis Zero, etc.
Target Customers	Existing soft drinks consumers and MZ trend setters who prefer low-calorie options
Business Strategy	Leverage mega-brand assets to rapidly penetrate the zero market and secure market share
Raw Material Sourcing	Allulose sourced from Group 4 (Samyang, Daesang) Beverage concentrate sourced from PepsiCo
Distribution	Strong nationwide retail network and convenience store distribution; concentrate supply system through PepsiCo partnership
Manufacturing & Production	Mass production and automated processes through large-scale smart factories in Anseong, Ochang, etc.
Marketing & Sales	Ad campaigns utilizing celebrities and influencers targeting people in their teens to 30s; pop-up stores, tasting events, and offline immersive campaigns
Organizational Structure & Personnel	Divided into beverage and mainstream business divisions; recently strengthening global business such as Philippine Pepsi to expand overseas revenue

Key Differentiators Within the Strategy Group

1	Keep	- Maintain mega-brand assets accumulated over decades (Chilsung Cider, Pepsi, etc.). - Leverage existing nationwide distribution network of 100,000+ partners. - Maintain cost advantage structure based on general-purpose sugar substitutes.
2	Eliminate	- Abandon separate premium pricing strategy in the zero beverage market; enter at the same price point as existing products → remove consumer resistance. - Minimize new manufacturing facility investment.
3	Create	Secure new Southeast Asian zero beverage distribution channels through Philippine Pepsi.

Changes in 5 Forces

- Bargaining Power of Buyers:** With a 49% share of the zero beverages market, consumer choices are narrowed, partially reducing bargaining power.
- Threat of New Entrants:** Triple barriers of scale, brand, and distribution effectively block small-scale new entrants.
- Intra-industry Rivalry:** Market share dominance effectively restructures the competitive landscape around Lotte Chilsung.

Leveraging strong existing brand recognition to maintain overwhelming distribution dominance

Group 2 Winner: Binggrae

Binggrae secured the #1 position in the ice cream market through the Haitai acquisition and is recording high profit margins with a broad steady-seller zero lineup.

Executive Summary

Item	Details
Zero Product Lineup	Melona Zero, Samanco Zero, Yoplait Zero (sugar-free), Powercap Zero, etc.
Target Customers	Health-conscious consumers of all ages interested in sugar intake and blood sugar management.
Business Strategy	Leverage mega-brand assets to rapidly penetrate the zero market and secure share. Dramatically improve operating profit through 100% acquisition of Haitai to catch up with competitor Lotte Wellfood.
Raw Material Sourcing	Sourcing from diverse suppliers. Raw materials procured directly from domestic farms and through the National Agricultural Cooperative Federation.
Distribution	Domestic distribution centered on convenience store channel. Prioritizing North American market penetration ahead of Melona — local sales through local subsidiaries in the US, China, Vietnam, and other markets.
Manufacturing & Production	Optimizing in-house production infrastructure. Pursuing efficiencies through logistics and production bas. integration with Haitai Ice Cream.
Marketing & Sales	Domestic and international MZ-generation-targeted rebranding and SNS viral marketing of mega brands.
Organizational Structure & Personnel	R&D organization comprised of refrigeration research, dairy research, fermentation & health research, beverage research, and food safety teams. Organizational streamlining through integration and voluntary retirement following the Haitai merger.

https://www.bing.co.kr/news/news_announced_view?anno_idx=242
<https://www.sme.daily.co.kr/news/articleView.html?idxno=353274>

Key Differentiators Within the Strategy Group

1	Keep	- Continue leveraging existing mega-brands (Melona, Yoplait) as the foundation of the zero lineup. - Maintain convenience store and large-format retail-centered offline distribution / Maintain and expand North American export channels.
2	Eliminate	Reduce low-profitability product lines and factories (based on announcements such as the sale of the Jungpyeong bread factory).
3	Create	- Pioneer K-dessert trends in the North American market and secure new local distribution channels overseas. - Build a new premium positioning through zero ice cream and dairy product lines.

Changes in 5 Forces

- **Intra-industry Rivalry:** Secured 42.93% ice cream market share through the Haitai merger (Buy strategy) → competitive intensity eased with restructured two-way rivalry against Lotte Wellfood (39.86%).
- **Threat of New Entrants:** Raising entry barriers through economies of scale and strengthened brand assets.
- **Bargaining Power of Buyers:** Rising consumer switching costs due to strengthened brand lock-in effect.

Maximizing zero product line profitability by leveraging mega-brand assets and manufacturing & export infrastructure

<https://www.ftoday.co.kr/news/articleView.html?idxno=357205>

Group 3 Winner: Lalasweet

Lalasweet pioneered the sugar-free dessert category and continues to achieve rapid growth through fandom marketing.

Executive Summary

Item	Details
Zero Product Lineup	Sugar-free chocolate bars, sugar-free ice cream, sugar-free popcorn, sugar-free macarons.
Target Customers	Diet- and sugar-conscious consumers in their 20s–30s. Consumers who want to enjoy guilty pleasures (high-calorie snacks) while reducing guilt.
Business Strategy	Differentiation strategy: maintaining a premium price point of approximately 3,000 KRW per unit.
Raw Material Sourcing	Blending sugar substitutes such as erythritol and allulose. outsourced production with no in-house factory (can act as a structural weakness in cost control).
Distribution	Blending sugar substitutes such as erythritol and allulose. Outsourced production with no in-house factory (can act as a structural weakness in cost control).
Manufacturing & Production	Leveraging SNS viral and fan-participatory marketing such as "Laladus Audition". Achieved #1 single-item ice cream sales ranking at CU convenience stores.
Marketing & Sales	Small asset-light structure. Operated solely on self-generated revenue without external investment since launch, achieving approximately KRW 60 billion in sales in 2024.

Key Differentiators Within the Strategy Group

1	Keep	• Maintain fandom-based SNS marketing structure. • Maintain premium pricing of approximately 3,000 KRW per unit. • Maintain self-revenue growth foundation without external investment.
2	Eliminate	• Remove cost structure of traditional offline distribution such as large-format mart entry (currently large marts are largely absent — also a channel limitation). • Refrain from excessive SKU expansion; maintain category-focused strategy.
3	Create	• Create the new category of "sugar-free ice cream" itself. Integrate fan-participatory new product development process (Laladus) into R&D. • Pioneer new overseas sugar-free dessert markets such as Vietnam.

Changes in 5 Forces

- **Bargaining Power of Buyers:** Consumer switching costs rise through fandom and category first-mover advantage, reducing bargaining power.
- **Threat of substitute products or services:** As the brand that first defined the category, large competitors' products function as substitutes rather than direct rivals.

Pioneering the sugar-free dessert market and securing strong brand loyalty

Group 4 Winner: Samyang

Samyang dominates the entire B2B and B2C value chain, leveraging the world's first allulose mass production technology and global licensing.

Executive Summary

Item	Details
Zero Product Lineup	B2B: Nexweet allulose/stevia, etc. B2C: Cure True Sweet allulose/stevia/allulose sugar, etc
Target Customers	B2B: Large domestic and international food manufacturers (Groups ①, ②) and startups (Group ③) B2C: MZ generation "healthpleasure" consumers and diabetes management customers who pursue both health and enjoyment
Business Strategy	Vertical integration and technological superiority strategy: produce raw materials using proprietary enzyme technology and apply them to subsidiary brand finished products, securing both cost competitiveness and quality reliability simultaneously.
Raw Material Sourcing	In-house production of core enzymes.
Distribution	B2B: Supplying to 300+ products domestically and internationally / B2C: Mart, online, and self-brand distribution Global: Global licensing and base establishment secured in China, US, Australia/New Zealand, etc.
Manufacturing & Production	Ulsan specialty factory in operation; secured largest domestic production line with KRW 140 billion investment in 2024; AI-based factory optimization applied.
Marketing & Sales	Providing customer-tailored solutions beyond simple raw material supply (AI-based sugar reduction solution released publicly) B2C emphasizes sugar-free expertise based on "Cure's" credibility
Item	Food Bio Research Institute conducts enzyme technology research and commercialization engineering; product development and applied research for diverse businesses, collaborating with business divisions

<https://www.insightkorea.co.kr/news/articleView.html?idxno=235211>
<https://www.bloter.net/news/articleView.html?idxno=658239>

Key Differentiators Within the Strategy Group

1	Keep	- Maintain world's first food-grade fermentation enzyme technology and dominant domestic market share. - B2B partnerships: Continue solid supply relationships with major domestic food companies such as Lotte Chilsung.
2	Eliminate	Reduce dependence on low-margin general-purpose ingredients; reduce proportion of low-profitability general-purpose ingredients such as regular sugar and shift to high-value-added allulose-centered portfolio improvement.
3	Create	- B2C brand awareness: Strengthen brand marketing to be recognized by consumers as a "sugar-free leader" beyond just a raw material supplier. - Global licensing first-mover advantage: Create new global legal sales qualifications that competitors cannot match

Changes in 5 Forces

- They made low forces of group 4 business structures even lower. Their large-scale facilities, patents, and global licenses made the entrance barrier thicker and bargaining power over buyer companies from other groups.

Dominating the value chain from B2B to B2C based on allulose patent technology and global licensing

<https://www.asiatoday.co.kr/kn/view.php?key=20240424010013733>

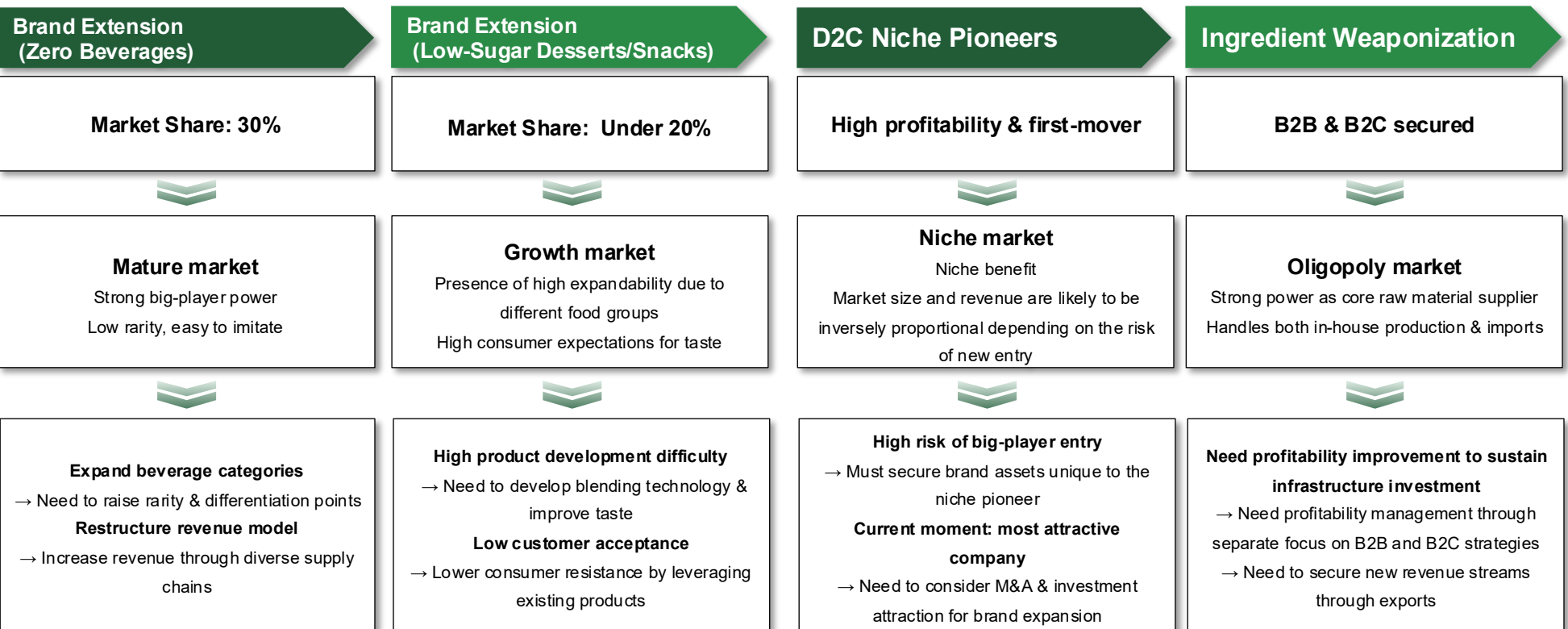
V. Strategic Recommendations by Group

1. Low-Sugar & Low-Calorie Industry Strategic Recommendations

Low-Sugar & Low-Calorie Industry Strategic Recommendations

Direction and necessity of Blue Ocean development for companies within the growing low-sugar & low-calorie food industry.

4 Strategic Groups



Key Interpretation

The era when "eating less" was seen as the answer → Found a '**blue ocean**' in the low-sugar & low-calorie industry through new alternative sweetener materials. However, even this space is now becoming increasingly competitive, gradually turning into a '**red ocean**' → Need to create a "**New Blue Ocean.**"



Thank You

Team D

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